Sia Zhi Hung

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Education

2026 - 2029 Physics (BSc) at **Imperial College London** (London, United Kingdom)
- First year

2018 - 2023 **Hwa Chong Institution** (Singapore)
- Singapore-Cambridge GCE A-Levels
- AAAA for H2 Physics, Chemistry, Mathematics and Economics (A is the highest grade)

Experience

Private Tutor 2026

Tutoring Singapore-Cambridge GCE A-Level H2 Physics.

Second Sergeant in the Singapore Artillery 2024 - 2025

Full-time military service (mandatory). Served as the platoon sergeant and led a group of 20 soldiers.

Vice-president — Hwa Chong String Ensemble 2022 - 2023

Coordinated practices, events and concerts for an orchestra of 30 students.

Volunteer Tutor 2021

A year-long non-profit tutoring programme. Held weekly sessions with 12-year-old students to cover all their academic subjects.

Awards

Bronze — Singapore Mathematical Olympiad (Open) 2023
Top 9% — Australian Mathematics Competition (Senior) 2022
Silver — Singapore Mathematical Olympiad (Senior) 2021

Projects

Double Pendulum Simulation and Chaos Map 2026

Derived the second-order differential equations of the double pendulum and integrated them using the fourth-order Runge-Kutta method to simulate its motion. Total energy conservation was used to verify accuracy. Made a colour-coded chaos map to show how small differences in the initial conditions cause the behaviour of different pendulums to diverge. Project is on GitHub (github.com/zhibungus/doublependulum).

Nash Equilibrium of Variations of the Public Goods Game 2020

Added complexity to the classic economic game by rewarding personal contributions, punishing free-riders and changing the win condition to investigate how the Nash equilibrium changed.

Skills and Interests

Python	Built various projects using NumPy, including dice roll games, geometric Brownian motion and the double pendulum simulator.
Modern Tetris	Formerly ranked 10th in Singapore, currently top 2.5% worldwide (Tetr.io). Organised and conducted a week-long Tetris course for my school in 2021.
Other	Microsoft Excel; bouldering